Student Performance Analysis

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1. Introduction

This project analyzes student performance based on various factors like gender, parental education, and exam scores.

2. Load Dataset

```
data <- read.csv("StudentsPerformance.csv")
head(data)
```

```
##   gender race.ethnicity parental.level.of.education        lunch
## 1 female        group B           bachelor's degree     standard
## 2 female        group C                some college     standard
## 3 female        group B             master's degree     standard
## 4   male        group A          associate's degree free/reduced
## 5   male        group C                some college     standard
## 6 female        group B          associate's degree     standard
##   test.preparation.course math.score reading.score writing.score
## 1                    none         72            72            74
## 2               completed         69            90            88
## 3                    none         90            95            93
## 4                    none         47            57            44
## 5                    none         76            78            75
## 6                    none         71            83            78
```

3. Data Cleaning

```
data <- na.omit(data)
str(data)
```

```
## 'data.frame':    1000 obs. of  8 variables:
##  $ gender                     : chr  "female" "female" "female"
"male" ...
##  $ race.ethnicity             : chr  "group B" "group C" "group B"
"group A" ...
##  $ parental.level.of.education: chr  "bachelor's degree" "some college"
"master's degree" "associate's degree" ...
##  $ lunch                      : chr  "standard" "standard" "standard"
"free/reduced" ...
##  $ test.preparation.course    : chr  "none" "completed" "none"
"none" ...
##  $ math.score                 : int  72 69 90 47 76 71 88 40 64 38 ...
##  $ reading.score              : int  72 90 95 57 78 83 95 43 64 60 ...
##  $ writing.score              : int  74 88 93 44 75 78 92 39 67 50 ...
```

4. Data Analysis

```
summary(data)
```

```
##        gender       race.ethnicity parental.level.of.education
lunch
##  Length   :1000   Length   :1000   Length   :1000              Length
:1000
##  N.unique :   2   N.unique :   5   N.unique :   6
N.unique :   2
##  N.blank  :   0   N.blank  :   0   N.blank  :   0
N.blank  :   0
##  Min.nchar:   4   Min.nchar:   7   Min.nchar:  11
Min.nchar:   8
##  Max.nchar:   6   Max.nchar:   7   Max.nchar:  18
Max.nchar:  12
##

##  test.preparation.course   math.score     reading.score
writing.score
##  Length   :1000          Min.   :  0.00   Min.   : 17.00   Min.   :
10.00
##  N.unique :   2          1st Qu.: 57.00   1st Qu.: 59.00   1st Qu.:
57.75
##  N.blank  :   0          Median : 66.00   Median : 70.00   Median :
69.00
##  Min.nchar:   4          Mean   : 66.09   Mean   : 69.17   Mean   :
68.05
##  Max.nchar:   9          3rd Qu.: 77.00   3rd Qu.: 79.00   3rd Qu.:
79.00
##                          Max.   :100.00   Max.   :100.00   Max.
:100.00
```

```
mean(data$math.score)
```

```
## [1] 66.089
```

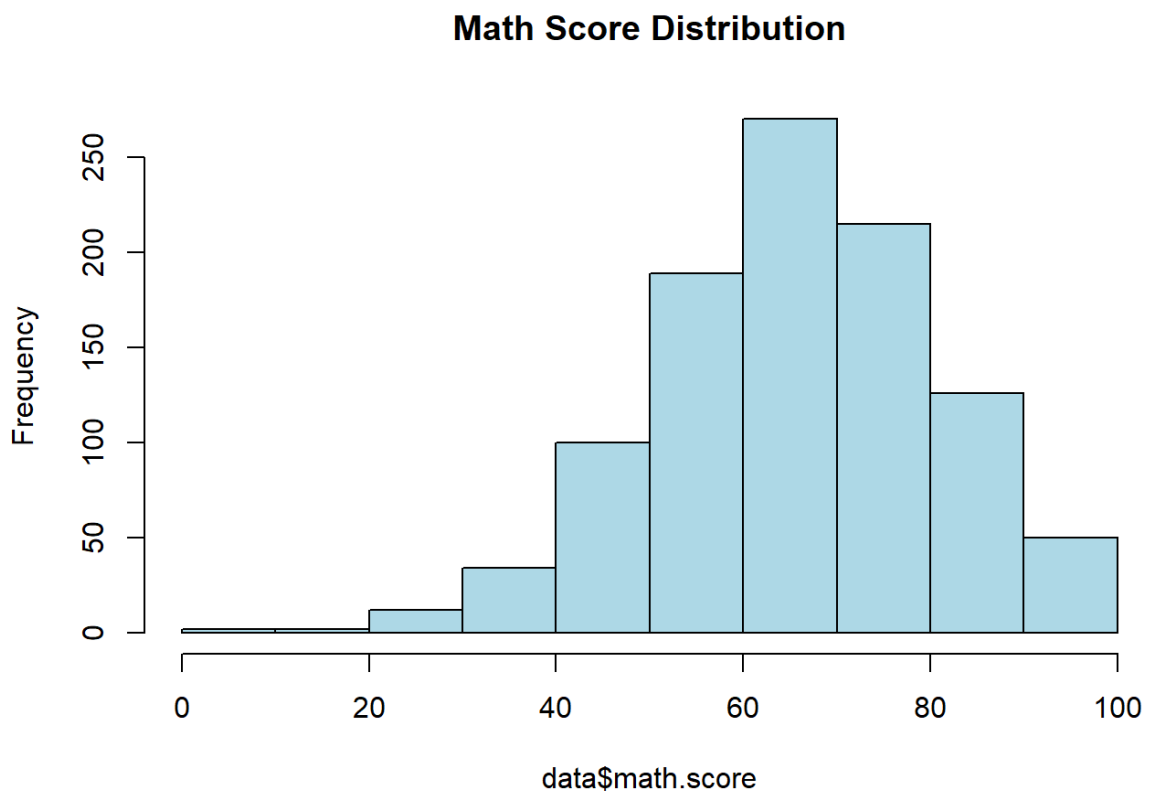
```
sd(data$math.score)
```

```
## [1] 15.16308
```

5. Graphs

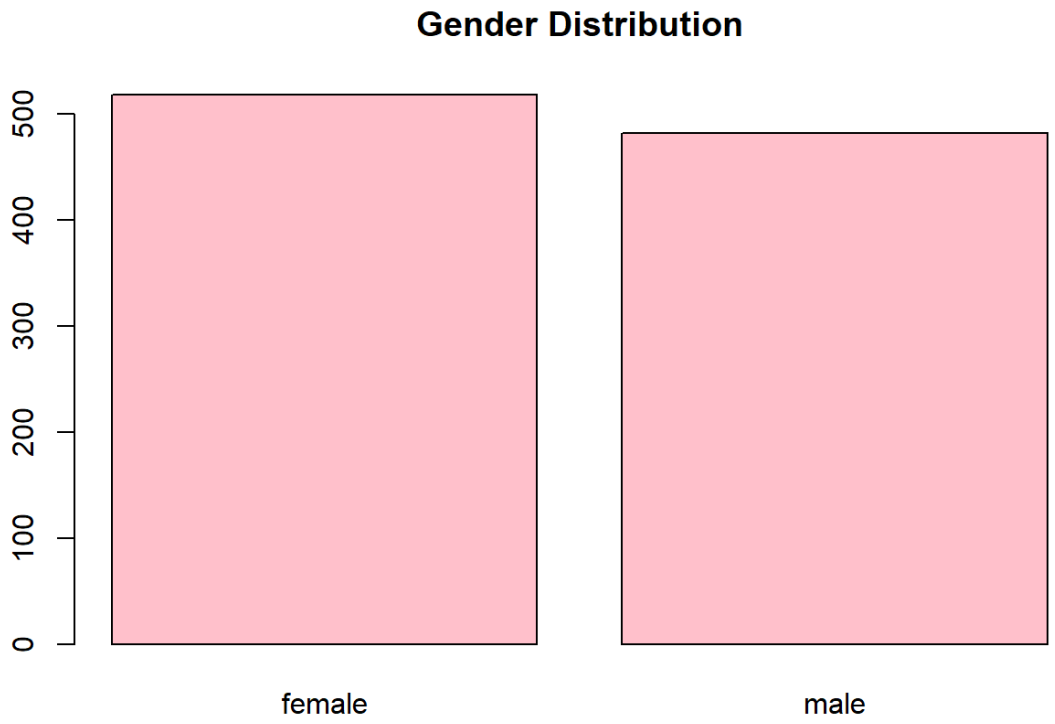
Histogram

```
hist(data$math.score, main="Math Score Distribution", col="lightblue")
```



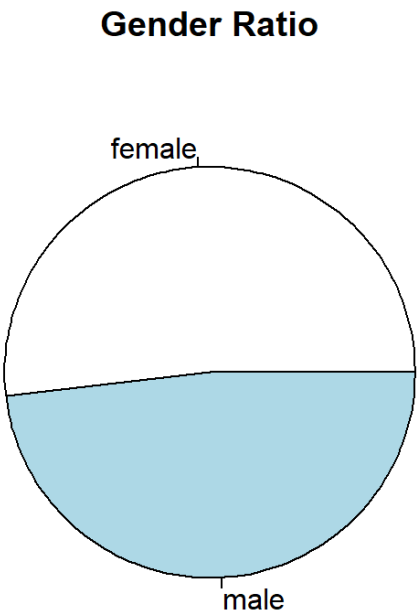
Bar Plot (Gender)

```
barplot(table(data$gender), main="Gender Distribution", col="pink")
```



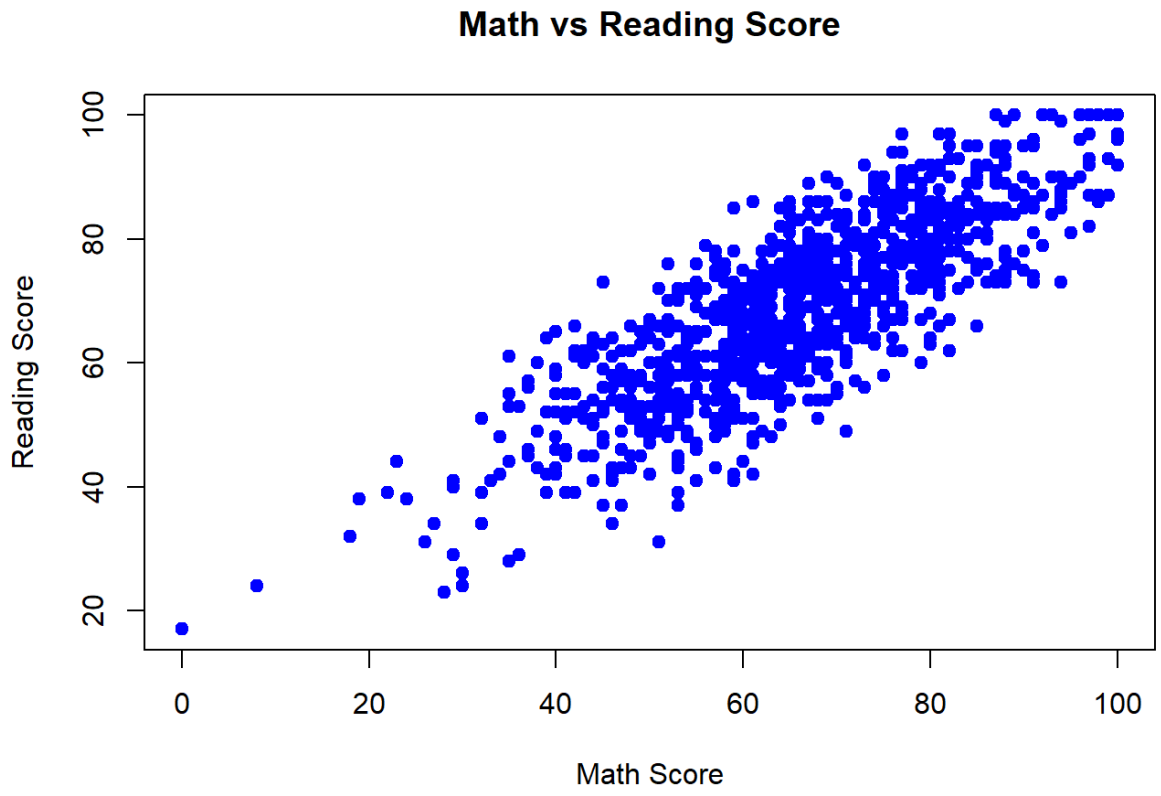
Pie Chart

```
pie(table(data$gender), main="Gender Ratio")
```



Scatter Plot

```
plot(data$math.score, data$reading.score,
     main="Math vs Reading Score",
     xlab="Math Score",
     ylab="Reading Score",
     col="blue", pch=19)
```



6. Correlation

```
cor(data$math.score, data$reading.score)
```

```
## [1] 0.8175797
```

7. Interpretation

The analysis shows relationships between different subject scores. Students with higher math scores tend to have higher reading scores.

8. Conclusion

Student performance is influenced by multiple factors, and strong relationships exist between subject scores.